

"We are in a phase where haptics is necessary,"

Designs and Projects Development has haptics-based products targeting surgery simulation, mission control and assembly operations. The products designed by it are sold through Merlyn D3 Systems.

Biju Ronnie Varkey, owner, Designs and Projects Development, speaks to Abhishek Mutha of EFY about haptics and challenges faced while designing advanced bomb-disposal bots



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What is haptics? Why is it gaining importance?

Today, we are in a phase where haptics is necessary. For example, when an arm in a robotic device is attached with a drilling tool to drill a suitcase with an explosive device, the amount of force required to drill through it is very important for a very safe disposal. The force applied by the robot has to be felt by the operator. Otherwise, there can be tremendous damage. If there is a haptic feedback, the operator knows exactly how the arm and the tool are operating and also the forces involved.

Secondly, when a trigger is being removed from an explosive device, it is important to know the pressure being applied by the gripper. Too much force might cause an explosion and too less will make it impossible to extricate the trigger. There should be a very precise control wherein the operator should feel that he is at the site performing all the operations. We are using this technology in surgery simulation too.

How does haptics work?

Take the simple example of riding a bicycle. We sense so many factors including the position of our body, the speed, the rumble of the road, the wind on our face, the exhilaration of gliding, the force we apply on the pedal and the effect it has on acceleration. Try doing that on a game console on your PC. You now know what you are missing.

According to medical science, there are more than 28 different types of sensors in our human body. These sensations can be used as a feedback mechanism to add to our feel of our action with their implementation in the machine being controlled by us.

Haptics is the science of feel for our perception senses so that our body can experience the result of our actions to control a robotic system. In a haptic system the end effector is attached to sensors to detect the process being undertaken. The force and orientation data is fed to the controlling device and the action mimicked so that the operator can feel the effect of the actions being given through the controlling device.

Traditionally, feedback was limited to the sense of force that is being applied by the end effector on the robotic system. So the force that is being applied is fed back to another robot actuator system that is integrated to the control device like a joystick. The feedback would be in a ratio as decided. It can be amplified if the force applied is very low so that the applied force perception is detectable. It can be lowered if the forces being applied by the robot are high.

How is the technology advancing?

As technology advances, more types of human perception are being used to function as a control feedback methodology in order to actively guide the robotic system. Sometimes these are also used in combination to achieve more complete involvement.

There are different ways of implementing a haptic device. The haptic device can have a feedback that is localised as the vibration of a mobile handset. It just intimates that your attention is needed and you have to look for it. It does not have any particular direction to point for your attention.

Some devices apply forces in a particular direction or vector. That is, they apply force in a 3D space of a particular size. These can be used to guide robotic arms in 3D space